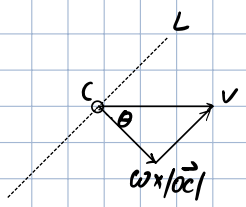
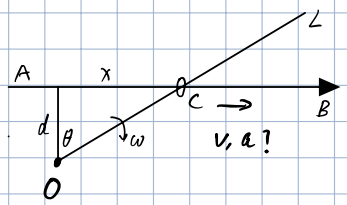
1.1

$|OC| = \sqrt{x^2+d^2}$



$\vec{i}$



$\vec{v} = \frac{\omega\sqrt{x^2+d^2}}{\cos\theta}\vec{i} = \frac{\omega(x^2+d^2)}{d}\vec{i}$

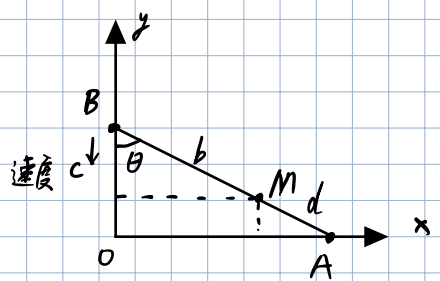
$\frac{d\vec{x}}{dt} = \frac{d(d\tan\theta)}{dt}\vec{i} = d\frac{d}{dt}\left(\frac{\sin\theta}{\cos\theta}\right)\vec{i}$

运动的合成与分解.

$= d\cdot\frac{1}{\cos^2\theta}\cdot\frac{d\theta}{dt}\vec{i} = \frac{d\omega}{\cos^2\theta}\vec{i} = \frac{d\omega(x^2+d^2)}{d^2} = \frac{x^2+d^2}{d}\omega\cdot\vec{i}$

$\vec{a} = \frac{d\vec{v}}{dt} = \frac{2\omega x}{d}\cdot\frac{d\vec{x}}{dt} = \frac{2\omega^2 x(x^2+d^2)}{d^2}\vec{i}$

1.2.



$y_B = (b+d)\cos\theta$

$\dot{y}_B = -(b+d)\sin\theta\cdot\dot{\theta} = c$

$\Leftrightarrow \dot{\theta} = -\frac{c}{(b+d)\sin\theta}$

$x_M = b\sin\theta;\quad y_M = d\cos\theta$

$\Rightarrow \dot{x}_M = b\cos\theta\cdot\dot{\theta} = -\frac{bc\cos\theta}{(b+d)\sin\theta}$

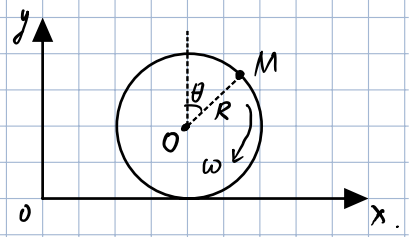
$\dot{y}_M = -d\sin\theta\cdot\dot{\theta} = \frac{cd}{b+d}$

$\Rightarrow \vec{v}_M = -\frac{bc}{b+d}\cot\theta\cdot\vec{i} + \frac{cd}{b+d}\vec{j}$

$|v_M| = \frac{c}{b+d}\sqrt{d^2+b^2\cdot\frac{\cos^2\theta}{\sin^2\theta}} = \frac{c}{b+d}\sqrt{d^2+b^2\cot^2\theta}$

$\vec{a}_M = -\frac{bc}{b+d}\frac{\dot{\theta}}{\sin^2\theta}\cdot\vec{i} + 0 = \frac{bc}{b+d}\frac{1}{\sin^2\theta}\cdot\frac{c}{(b+d)\sin\theta}\vec{i} = \frac{bc^2}{(b+d)^2}\frac{1}{\sin^3\theta}\vec{i}$

1.3



设 O 点坐标 $(\omega Rt + x_0, R)$

则：M 点坐标：$(\omega Rt + x_0 + R\sin\theta,\ R + R\cos\theta)$

$$\vec{v}_M = (\omega R + R\cos\theta \cdot \dot{\theta})\vec{i} + R(-\sin\theta \cdot \dot{\theta})\vec{j}$$

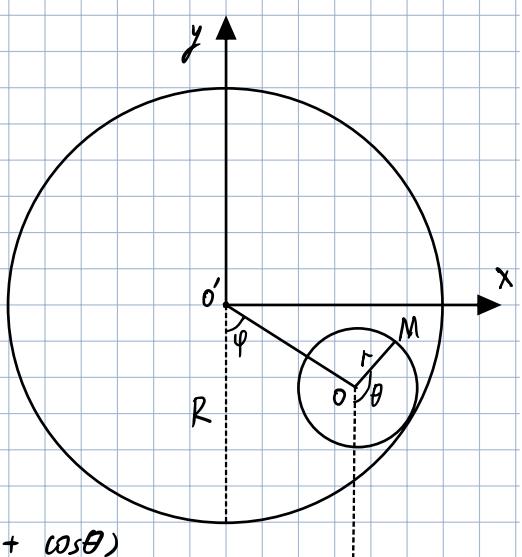
$$= \omega R(1+\cos\theta)\vec{i} + (-R\omega\sin\theta)\vec{j}.$$

$$\vec{a}_M = (-\omega R\sin\theta \cdot \dot{\theta})\vec{i} + (-\omega\cos\theta \cdot \dot{\theta})\vec{j} = -\omega^2 R(\sin\theta\vec{i} + \cos\theta\vec{j})$$

$$= \omega^2 R\vec{e}_r$$

没有必要觉得为难，先把坐标写出来求导即可.

1.4 方法一：



$\dot{\theta} = \omega$

$\dot{\varphi} = -\omega r/R$.

O 坐标 $(R\sin\varphi, -R\cos\varphi)$

$M(R\sin\varphi + r\sin\theta,\ -R\cos\varphi - r\cos\theta)$

$$\dot{x}_M = R\cos\varphi \cdot \dot{\varphi} + r\cos\theta \cdot \dot{\theta} = \omega r(-\cos\varphi + \cos\theta)$$

$$\dot{y}_M = R\sin\varphi \cdot \dot{\varphi} + r\sin\theta \cdot \dot{\theta} = \omega r(-\sin\varphi + \sin\theta)$$

$$\ddot{x}_M = -\frac{r^2\omega^2}{R-r}\sin\varphi - r\omega^2\sin\theta$$

$$\ddot{y}_M = \frac{r^2\omega^2}{R-r}\cos\varphi + r\omega^2\cos\theta.$$

$$\vec{v} = \dot{x}\sin\varphi\vec{e}_R + \dot{x}\cos\varphi\vec{e}_\varphi - \dot{y}\cos\varphi\vec{e}_R + \dot{y}\sin\varphi\vec{e}_\varphi$$

$$= (-r\omega\cos\varphi + r\omega\cos\theta)\sin\varphi\vec{e}_R + (-r\omega\cos\varphi + r\omega\cos\theta)\cos\varphi\vec{e}_\varphi$$
$$-(-r\omega\sin\varphi + r\omega\sin\theta)\cos\varphi\vec{e}_R + (-r\omega\sin\varphi + r\omega\sin\theta)\sin\varphi\vec{e}_\varphi$$

$$= -r\omega\cos\varphi\sin\varphi\vec{e}_R + r\omega\cos\theta\sin\varphi\vec{e}_R - r\omega\cos^2\varphi\vec{e}_\varphi + r\omega\cos\theta\cos\varphi\vec{e}_\varphi$$
$$+ r\omega\sin\varphi\cos\varphi\vec{e}_R - r\omega\sin\theta\cos\varphi\vec{e}_R - r\omega\sin^2\varphi\vec{e}_\varphi + r\omega\sin\theta\sin\varphi\vec{e}_\varphi$$

$$= r\omega\sin(\varphi-\theta)\vec{e}_R - r\omega\vec{e}_\varphi + r\omega\cos(\theta-\varphi)\vec{e}_\varphi$$

$$\vec{a} = \ddot{x}\sin\varphi\vec{e}_R + \ddot{x}\cos\varphi\vec{e}_\varphi - \ddot{y}\cos\varphi\vec{e}_R + \ddot{y}\sin\varphi\vec{e}_\varphi$$

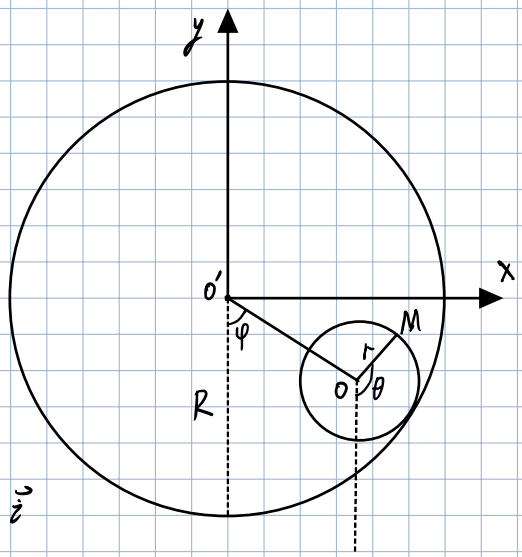
$$= \left(-\frac{r^2\omega^2}{R-r}\sin\varphi - r\omega^2\sin\theta\right)\sin\varphi\vec{e}_R + \left(-\frac{r^2\omega^2}{R-r}\sin\varphi - r\omega^2\sin\theta\right)\cos\varphi\vec{e}_\varphi$$
$$-\left(\frac{r^2\omega^2}{R-r}\cos\varphi + r\omega^2\cos\theta\right)\cos\varphi\vec{e}_R + \left(\frac{r^2\omega^2}{R-r}\cos\varphi + r\omega^2\cos\theta\right)\sin\varphi\vec{e}_\varphi$$

$$= -\frac{r^2\omega^2}{R-r}\sin^2\varphi\vec{e}_R - r\omega^2\sin\theta\sin\varphi\vec{e}_R - \frac{r^2\omega^2}{R-r}\sin\varphi\cos\varphi\vec{e}_\varphi - r\omega^2\sin\theta\cos\varphi\vec{e}_\varphi$$
$$-\frac{r^2\omega^2}{R-r}\cos^2\varphi\vec{e}_\varphi - r\omega^2\cos\theta\cos\varphi\vec{e}_R + \frac{r^2\omega^2}{R-r}\cos\varphi\sin\varphi\vec{e}_\varphi + r\omega^2\cos\theta\sin\varphi\vec{e}_\varphi$$

$$= -\frac{r^2\omega^2}{R-r}\vec{e}_R - r\omega^2\cos(\theta-\varphi)\vec{e}_R - r\omega^2\sin(\theta-\varphi)\vec{e}_\varphi$$

$\vec{e}_R$, $\vec{e}_\varphi$ 方向？

方法二. 速度叠加



$$\vec{r} = r\sin\theta\,\vec{i} + r\sin(\theta-\tfrac{\pi}{2})\,\vec{j}$$

$$\vec{v} = \vec{v}_0 + \vec{\omega}\times\vec{r}$$

$$= (R-r)\dot{\varphi}\cos\varphi\,\vec{i} + (R-r)\dot{\varphi}\sin\varphi\,\vec{j} + \omega\vec{k}\times[r\sin\theta\,\vec{i} + r\sin(\theta-\tfrac{\pi}{2})\,\vec{j}]$$

$$= -r\omega\cos\varphi\,\vec{i} - r\omega\sin\varphi\,\vec{j} + r\omega\sin\theta\,\vec{j} + r\omega\cos\theta\,\vec{i}$$

$$R\dot{\varphi} = -r(\dot{\theta}-\dot{\varphi})$$

$$\dot{\varphi} = -\frac{r}{R-r}\dot{\theta}$$

$$\vec{a} = \vec{a}' + \vec{a}_0 + \dot{\vec{\omega}}\times\vec{r} + \vec{\omega}\times(\vec{\omega}\times\vec{r}) + 2\vec{\omega}\times\vec{v}'$$

$$= \dot{\varphi}^2(R-r)[\cos(\tfrac{\pi}{2}+\varphi)\,\vec{i} + \cos\varphi\,\vec{j}] + \omega\vec{k}\times(r\omega\cos\theta\,\vec{i} + r\omega\sin\theta\,\vec{j})$$

$$= \frac{r^2\omega^2}{R-r}(-\sin\varphi\,\vec{i} + \cos\varphi\,\vec{j}) + r\omega^2\cos\theta\,\vec{j} - r\omega^2\sin\theta\,\vec{j}$$

1.5.

$$y = r\sin\theta = bt \qquad \theta = ct, \quad t = \frac{\theta}{c}$$

$$r\sin\theta = \frac{b\theta}{c} \iff cr\sin\theta = b\theta, \quad r = \frac{b\theta}{c\sin\theta}$$

$$\dot{r} = \frac{bc\sin\theta - \cos\theta\, b\theta}{c^2\sin^2\theta}\dot{\theta} = \frac{b}{\sin\theta}(1-\theta\cot\theta)$$

$$\vec{v} = \dot{\vec{r}} = \frac{b}{\sin\theta}(1-\theta\cot\theta)\vec{e}_r + \frac{b\theta}{\sin\theta}\vec{e}_\theta = \frac{b}{\sin\theta}[(1-\theta\cot\theta)\vec{e}_r + \theta\vec{e}_\theta]$$

1.6.

$$\vec{r} = r\vec{e}_r \qquad \dot{\vec{r}} = \dot{r}\vec{e}_r + \dot{\theta} r \vec{e}_\theta = \lambda r \vec{e}_r + \mu\theta \vec{e}_\theta$$

$$\therefore \dot{r} = \lambda r \;;\; \dot{\theta} r = \mu\theta. \iff \dot{r} = \lambda r \qquad \dot{\theta} = \frac{\mu\theta}{r}$$

$$\vec{a} = \ddot{\vec{r}} = \lambda \dot{r}\vec{e}_r + \lambda r \dot{\theta}\vec{e}_\theta + \mu\dot{\theta}\vec{e}_\theta - \mu\theta r\dot{\theta}\vec{e}_r$$

$$= \lambda \dot{r}\vec{e}_r + \lambda r\dot{\theta}\vec{e}_\theta + \mu\dot{\theta}\vec{e}_\theta - \mu\theta\dot{\theta}\vec{e}_r$$

$$= \lambda^2 r\vec{e}_r + \lambda\mu\theta\vec{e}_\theta + \frac{\mu^2\theta}{r}\vec{e}_\theta - \frac{\mu^2\theta^2}{r}\vec{e}_r$$

$$= \left(\lambda^2 r - \frac{\mu^2\theta^2}{r}\right)\vec{e}_r + \left(\lambda\mu\theta + \frac{\mu\theta^2}{r}\right)\vec{e}_\theta$$

1.7. $\vec{v} = v_x\vec{i} + v_y\vec{j} \qquad |\vec{v}| = v = \sqrt{v_x^2 + v_y^2} = \text{const}.$

$$\frac{d\sqrt{v_x^2+v_y^2}}{dt} = \frac{1}{2\sqrt{v_x^2+v_y^2}}(2v_x a_x + 2v_y a_y) = 0$$

$$\iff v_x a_x + v_y a_y = 0. \quad \text{即 } \vec{a} = (a_x, a_y) \perp \vec{v} = (v_x, v_y).$$

1.8. $r = k(1+\cos\theta) \qquad \dot{r} = -k\sin\theta\cdot\dot{\theta} \qquad (k^2\sin^2\theta + 1)\dot{\theta}^2 = v^2$

$$\dot{\theta} = \frac{1}{\sqrt{1+k^2\sin^2\theta}}v \qquad \dot{r} = -\frac{vk\sin\theta}{\sqrt{1+k^2\sin^2\theta}}$$

$$\vec{v} = \dot{r}\vec{e}_r + r\dot{\theta}\vec{e}_\theta = -\frac{vk\sin\theta}{\sqrt{1+k^2\sin^2\theta}}\vec{e}_r + \frac{vk(1+\cos\theta)}{\sqrt{1+k^2\sin^2\theta}}\vec{e}_\theta$$

附加：

1-1 已知一质点运动，径向与横向运动速度分别为 λr 与 $\mu\theta$，(λ 与 μ 为常数) 求质点加速度.

$$\vec{r} = r\vec{e}_r$$

$$\vec{v} = (r\vec{e}_r)' = \dot{r}\vec{e}_r + r\cdot\dot{\vec{e}}_r = \dot{r}\vec{e}_r + r\dot{\theta}\vec{e}_\theta = \lambda r\vec{e}_r + \mu\theta\vec{e}_\theta$$

$$\Leftrightarrow \dot{r} = \lambda r \quad \dot{\theta} = \mu\theta \quad \Rightarrow \ddot{r} = \lambda^2 r, \quad \ddot{\theta} = \mu^2\theta$$

$$\vec{a} = \dot{\vec{v}} = \lambda\dot{r}\vec{e}_r + \lambda r\dot{\vec{e}}_r + \mu\dot{\theta}\vec{e}_\theta + \mu\theta\dot{\vec{e}}_\theta$$

$$= \lambda\dot{r}\vec{e}_r + \lambda r\dot{\theta}\vec{e}_\theta + \mu\dot{\theta}\vec{e}_\theta - \mu\theta\dot{\theta}\vec{e}_\theta$$

$$= \left(\lambda^2 r - \frac{\mu^2\theta^2}{r}\right)\vec{e}_r + \left(\lambda\mu\theta + \frac{\mu^2\theta}{r}\right)\vec{e}_\theta$$

1-2. 一质点沿心脏线 $r=k(1+\cos\theta)$ 以恒定速率 v 运动，求质点的速度与加速度.

解：

$$\vec{r}=r\vec{e}_r$$

$$\vec{v}=\dot{\vec{r}}=\dot{r}\vec{e}_r+r\dot{\vec{e}}_r=\dot{r}\vec{e}_r+r\dot{\theta}\vec{e}_\theta$$

$$r=k(1+\cos\theta)\qquad \dot{r}=-k\sin\theta\cdot\dot{\theta}$$

$$v^2=\dot{r}^2+r^2\dot{\theta}^2=+k^2\sin^2\theta\cdot\dot{\theta}^2+k^2(1+\cos\theta)^2\dot{\theta}^2=k^2\dot{\theta}^2(2+2\cos\theta)$$

$$\Leftrightarrow\ \dot{\theta}=\pm\sqrt{\frac{v^2}{2k^2(1+\cos\theta)}}\qquad \dot{r}=\mp k\sin\theta\cdot\sqrt{\frac{v^2}{2k^2(1+\cos\theta)}}=\mp\sqrt{\frac{v^2\sin^2\theta}{2(1+\cos\theta)}}$$

$$\vec{v}=\mp\sqrt{\frac{v^2\sin^2\theta}{2(1+\cos\theta)}}\vec{e}_r\pm r\sqrt{\frac{v^2}{2k^2(1+\cos\theta)}}\vec{e}_\theta$$

$$=\mp\frac{v\sin\theta}{\sqrt{2(1+\cos\theta)}}\vec{e}_r\pm\sqrt{\frac{1+\cos\theta}{2}}v\vec{e}_\theta$$

$$\vec{a}=-\frac{3v^2}{4k}\vec{e}_r-\frac{2\sin\theta+\sin\theta\cos\theta}{2+2\cos\theta}\frac{v^2}{k}\vec{e}_\theta$$

$$\dot{\vec{e}}_r=\dot{\theta}\vec{e}_\theta\ ;\quad \dot{\vec{e}}_\theta=-\dot{\theta}\vec{e}_r.$$

1-3 将一质点 m 以初速度 v_0 与水平线成 α 角度抛出，此质点受到的空气阻力是其速率的 mk 倍（$k>0$ 为常数）证明：当质点的速度方向又与水平线成 α 角时，所需时间为？

解：

$$\begin{cases} \dfrac{dv_\perp}{dt} = -g - kv_\perp & v_\perp|_{t=0} = v_0\sin\alpha \\ \dfrac{dv_{//}}{dt} = -kv_{//} & v_{//}|_{t=0} = v_0\cos\alpha \end{cases}$$

（解微分方程）

① 对 $kv_\perp + \dot{v}_\perp = -g$，设 $v_\perp = a_1 e^{b_1 t} + c_1$

$$c_1 k + k a_1 e^{b_1 t} + a_1 b_1 e^{b_1 t} = -g \Rightarrow \begin{cases} k + b_1 = 0, & b_1 = -k \\ c_1 k = g, & c_1 = -\dfrac{g}{k} \end{cases}$$

$$v_\perp = a_1 e^{-kt} - \frac{g}{k}, \quad v_\perp|_{t=0} = a_1 - \frac{g}{k} = v_0\sin\alpha$$

$$\therefore \underline{v_\perp = \left(v_0\sin\alpha + \frac{g}{k}\right)e^{-kt} - \frac{g}{k}}$$

② 对 $\dot{v}_{//} = -kv_{//}$ 设 $v_{//} = a_2 e^{b_2 t} + c_2$

$$a_2 b_2 e^{b_2 t} = -k(a_2 e^{b_2 t} + c_2) \Longleftrightarrow (k + b_2) a_2 e^{b_2 t} + k c_2 = 0$$

$$\Longleftrightarrow c_2 = 0, \quad b_2 = -k, \quad v_{//} = a_2 e^{-kt}$$

$$v_{//}|_{t=0} = a_2 = v\cos\alpha, \quad \underline{v_{//} = v\cos\alpha\, e^{-kt}}$$

③ $$\frac{v_\perp}{v_{//}} = \frac{\left(v_0\sin\alpha + \frac{g}{k}\right)e^{-kt} - \frac{g}{k}}{v\cos\alpha\, e^{-kt}} = -\operatorname{tg}\alpha$$

$$\left(v_0\sin\alpha + \frac{g}{k} + v\cos\alpha\operatorname{tg}\alpha\right)e^{-kt} = \frac{g}{k}$$

$$\Leftrightarrow e^{-kt} = \frac{\frac{g}{k}}{2v_0\sin\alpha + \frac{g}{k}} = \frac{g}{2v_0 k\sin\alpha + g}$$

$$t = -\frac{1}{k}\ln\frac{g}{2v_0 k\sin\alpha + g} = \frac{1}{k}\ln\left(1 + \frac{2v_0 k\sin\alpha}{g}\right)$$

通过解微分方程求解问题.

1-4 一质点在势能函数为 $V(x)=V_0\left(\frac{a}{x}+\frac{x}{a}\right)$ 的力场中的 $x>0$ 的区域内运动. 此时 $V_0 a>0$, 求出稳定平衡点的位置, 并求出在这些点附近作微振动的频率.

$$\frac{\partial V}{\partial x}=V_0\left(-\frac{a}{x^2}+\frac{1}{a}\right)=0 \iff x=|a|.$$

$$\left.\frac{\partial^2 V}{\partial x^2}\right|_{x=|a|}=2V_0\frac{a}{x^3}=2\frac{V_0}{a^2}>0. \quad \therefore \text{是稳定平衡点}$$

$$T=2\pi\sqrt{\frac{m}{V_0''}} \iff f=\frac{1}{2\pi}\sqrt{\frac{2|V_0|}{a^2 m}}$$

答案为: $f=2\pi\sqrt{\frac{2|V_0|}{a^2 m}}$?